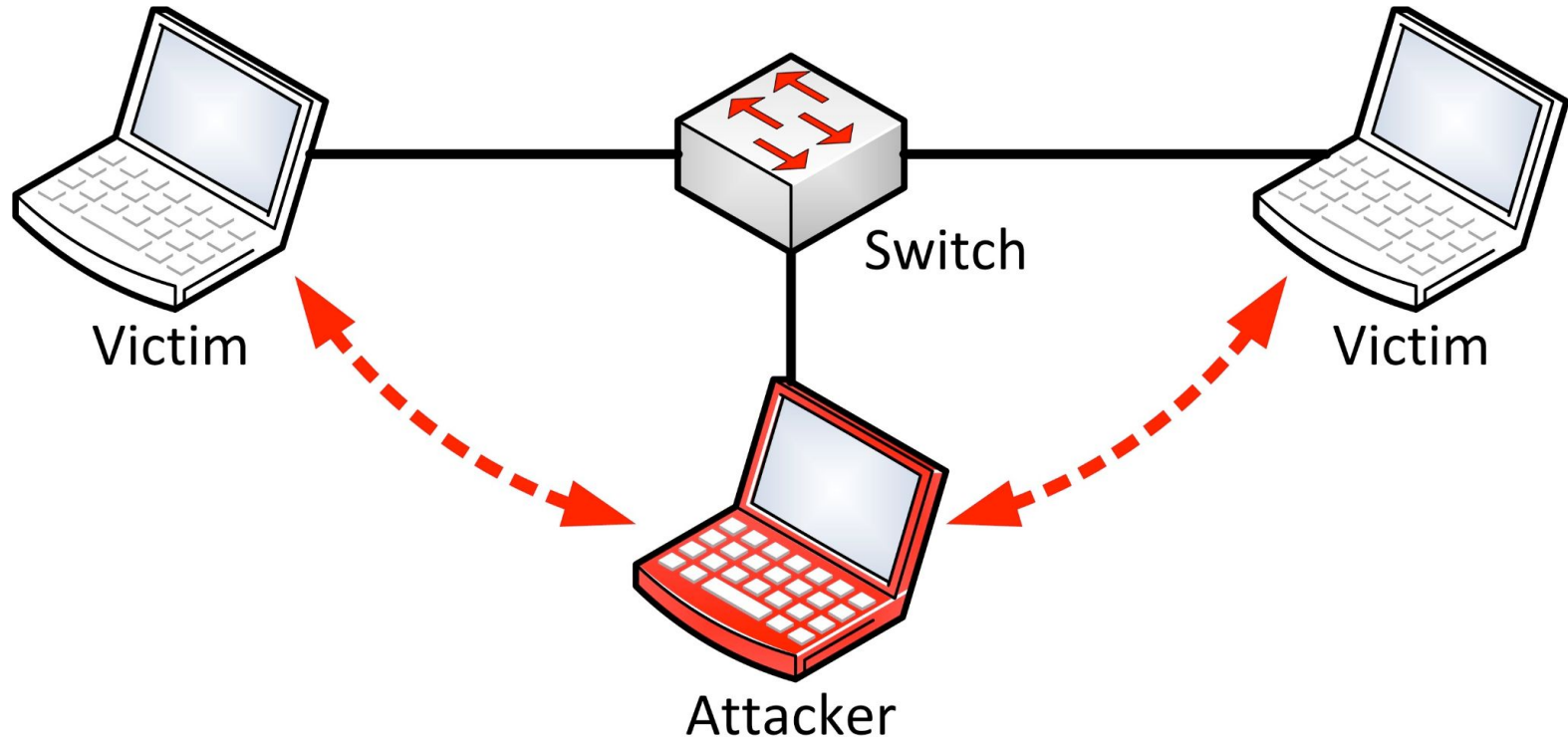


USB MITM Sniffing for Keylogging

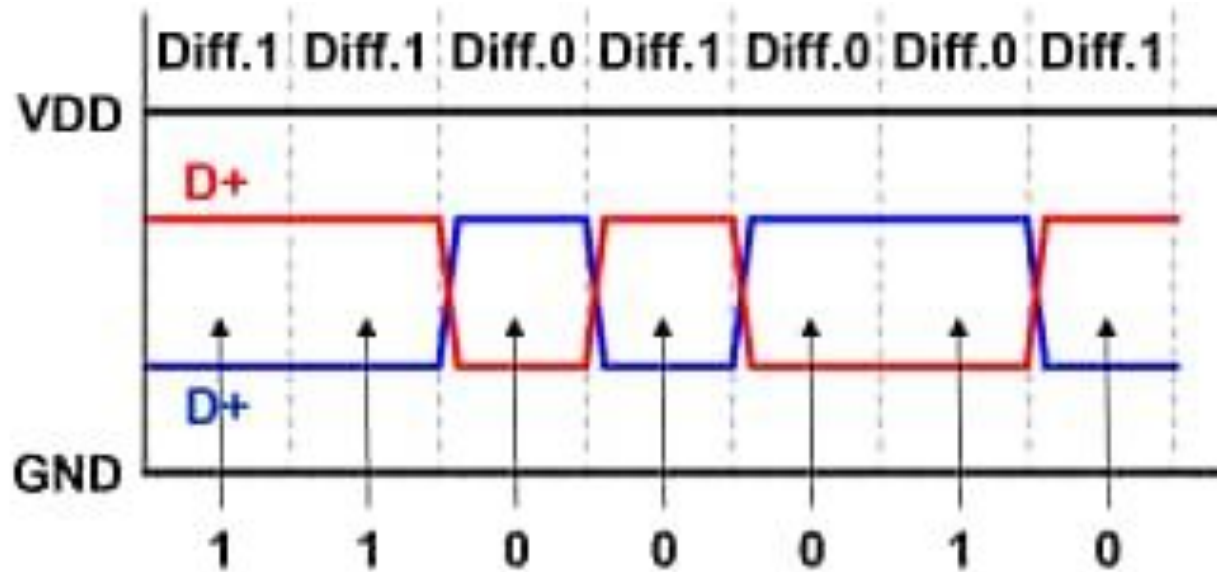


UW Cybersecurity

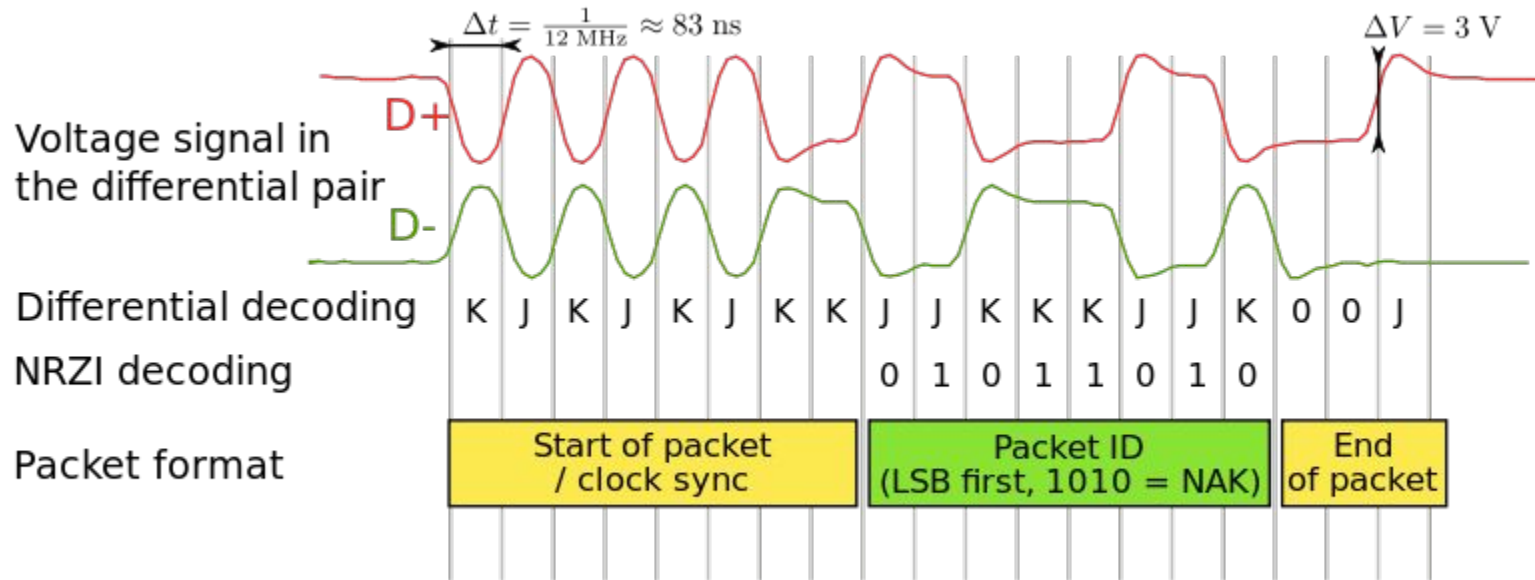
What is MITM?



What is USB?

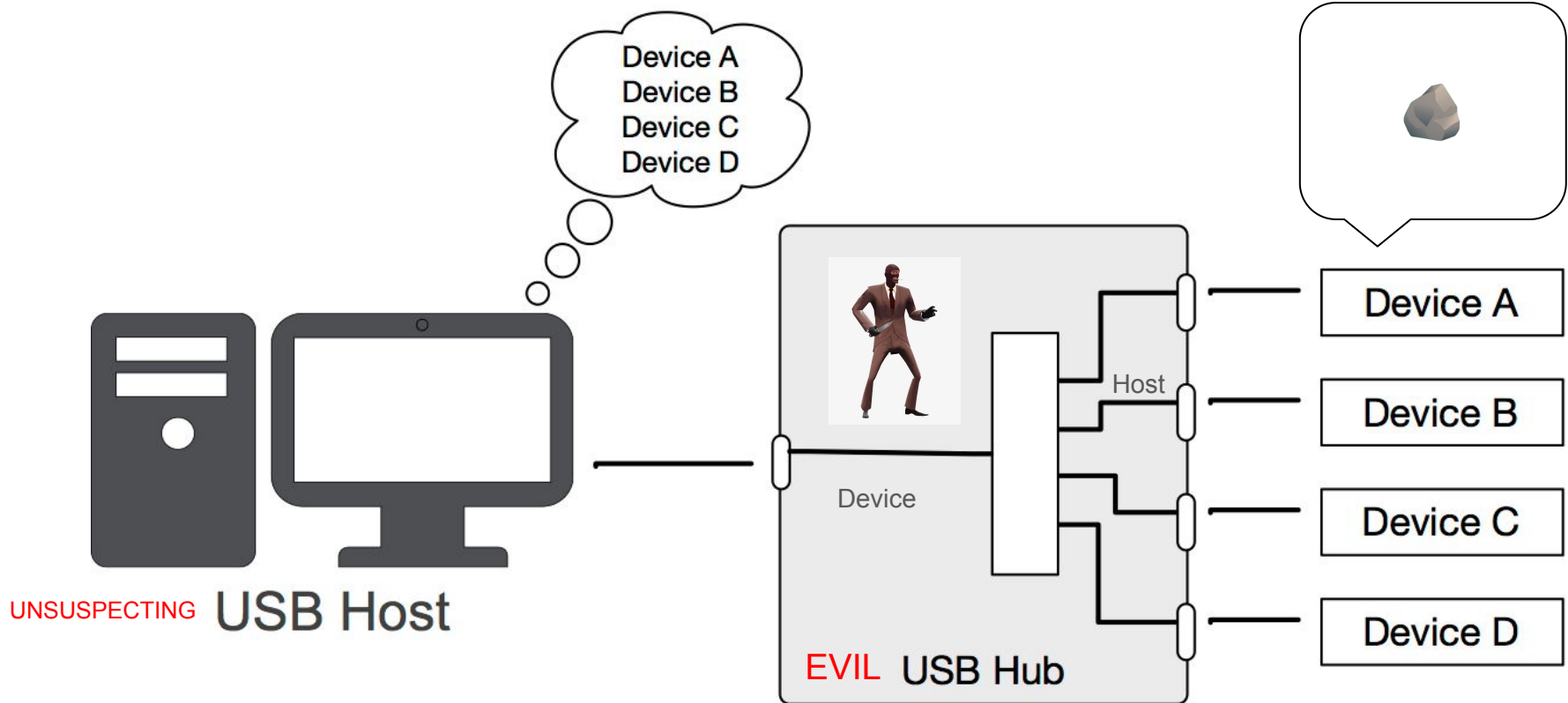


What is USB?

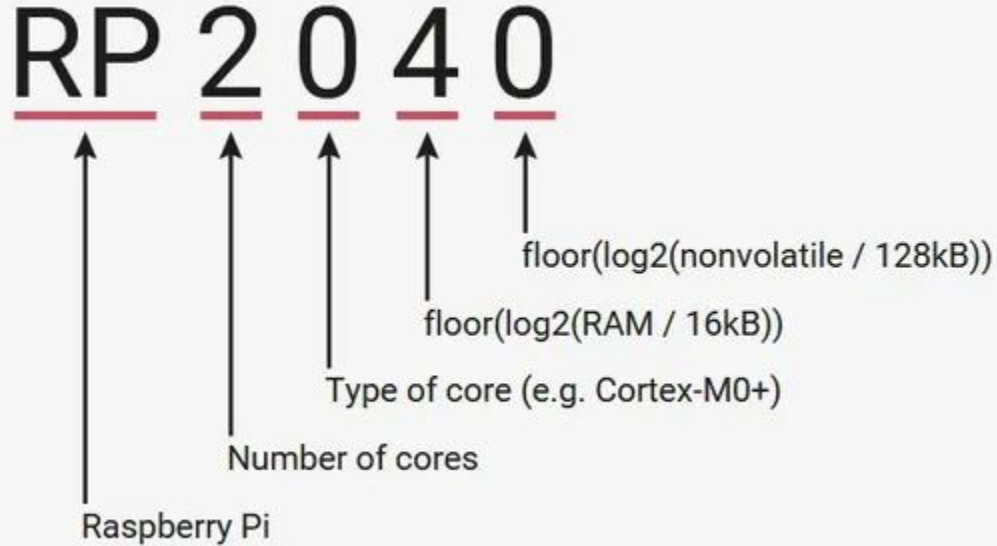


Hmmmm... This seems important....

What's a USB host vs Device?



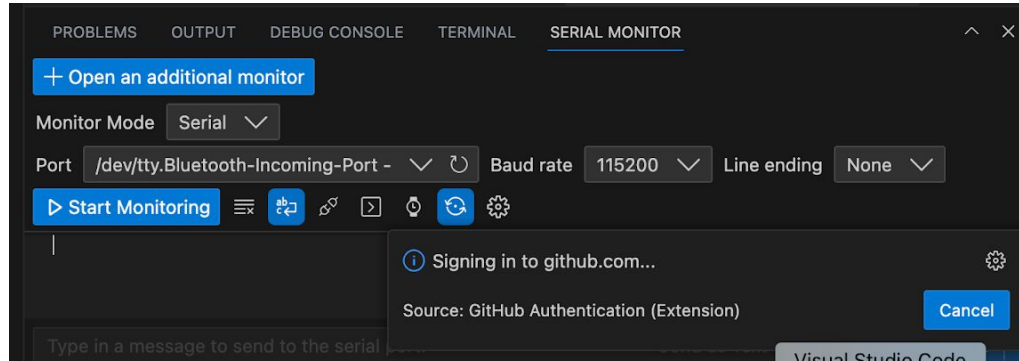
RP2040



How do embedded systems log information?



How do we talk to embedded systems?

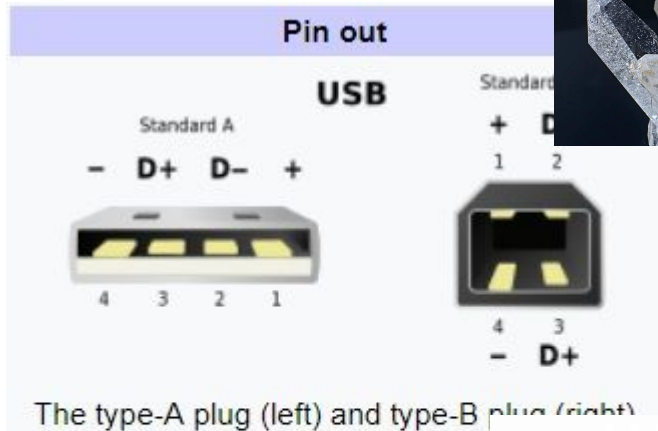


Protocol	UART	SPI	I2C
Complexity	Simple	Complex as device increases	Easy to chain multiple devices
Speed	Slowest	Fastest	Faster than UART
Number of devices	Up to 2 devices	Many, but gets complex	Up to 127, but gets complex
Number of wires	2	4	2
Duplex	Full Duplex	Full Duplex	Half Duplex
No. of masters and slaves	Single to Single	1 master, multiple slaves	Multiple slaves and masters

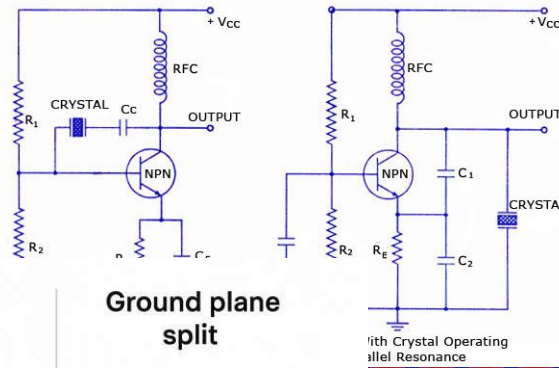
A side tangent into PCB design



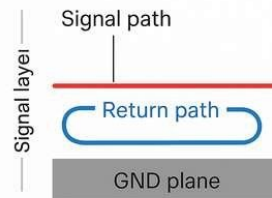
USB mistakes



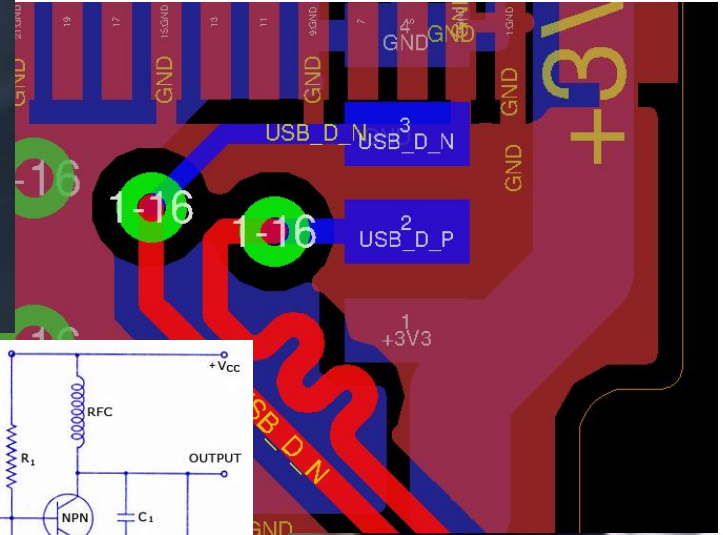
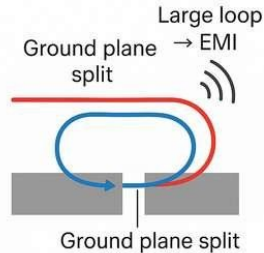
Pin 1	■ V_{BUS} (+5 V)
Pin 2	□ Data-
Pin 3	■ Data+
Pin 4	■ Ground



Continuous ground plane



Ground plane split



Demo time!

- Demo time!
- Demo time!
- Demo time!